

Chapter 6

Circles, Part 1 Of 2

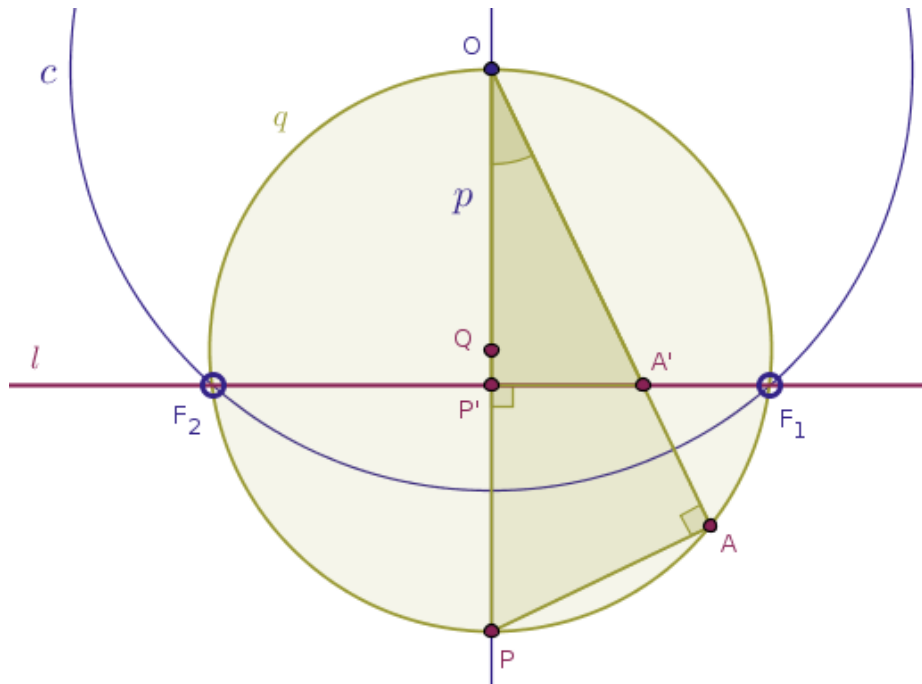
In this chapter we prove a theorem that captures a fundamental property of inversive geometry and which, in some sense, can be called The Fundamental Theorem Of Inversive Geometry, if we were to model the name of such a theorem after similar theorems in other layers of mathematics: The Fundamental Theorem Of Arithmetic in the Theory Of Numbers, the Fundamental Theorem Of Algebra about the roots of certain polynomials, The Fundamental Theorem Of Calculus in Calculus, The Fundamental Theorem On Group Homomorphisms (due to Emmy Noether) in the Theory of Groups and so on.

But first we summarize the straightforward results that are the converses of the theorems that we proved for straight lines in the previous chapter.

From the proofs of the five theorems that we generated in Chapter 5, Theorem 10, 11, 12, 13 and 14, we see that, for the most part, the inversion of the Euclidean plane with respect to a circle with either power, positive and negative, transforms a straight line into a circle that passes through the center of inversion.

It should also be clear that the converse statements of the above theorems are also true. With one exception of a sample proof, below we will only formulate these theorems for the record but we will omit their complete proofs since they all quite mechanically follow the same generic, templatizable, pattern:

- draw a straight line, p , of centers of the two given circles or a straight line that passes through the center of inversion, O , and the center, Q , of the given circle, q , until that straight line, p , intersects the given circle, q , at a point P , see the diagram in Figure 6.1 below
- invert that point P with the needed power into its image point, P'
- choose a point $A \neq P$ on the given circle q at random
- invert that point A with the needed power into its image point A'
- form the relevant triangles (Figure 6.1):



- use the main inversion formula from Condition 3 of Definition 1 in order to prove that these triangles, of which one is right by construction, are similar and, hence, the remaining triangle is right also
- thus, show that the locus of the image points A' will be a side of a right triangle or a straight line, which is what is require to prove, etc.

As such.

The following theorem is the converse of the Theorem 10.

Theorem 15. Under the inversion of a plane with respect to a circle $c(O, r)$ with positive power a circle q that passes through the center of inversion, O , and has exactly two distinct points, F_1 and F_2 , common with the circle of inversion, c , is transformed into a straight line, l , that passes through the said common points, F_1 and F_2 .

Proof. As a sample proof that can be straightforwardly adjusted for the remaining four theorems, we, first, draw a straight line, p , that passes through the center of inversion, O , and the center, Q , of the given circle, q . That straight line, p , is also known as **the straight line of centers** of these two circles.

The straight line p will intersect the given circle, q , at two distinct points, the center of inversion, O , and another point, that we will name P , that will be located strictly outside of the circle of inversion, c .

By the Theorem 5, under the inversion of the parent plane with respect to the given circle $c(O, r)$ with positive power that point P will be taken into its image point P' that will be located strictly inside of that circle of inversion.

On the given circle, q , pick a point, A , that is distinct from the point P and reflect the point A in the circle of inversion with positive power into its image point A' .

Connect the point A with the center of inversion, O , and the point P . Since the resultant angle OAP subtends the diameter, OP , of the circle of inversion, c , it follows that that angle must be right:

$$\angle OAP = \frac{\pi}{2}$$

Consider the triangle $OP'A'$. The interior angle at O of that triangle is shared with the same angle of the right triangle OAP :

$$\angle P'OA' = \angle POA$$

By the Condition 3 of Definition 1, we have for the points P and P' :

$$|OP| \times |OP'| = r^2$$

and for the points A and A' :

$$|OA| \times |OA'| = r^2$$

Since the radius of inversion, r , is one and the same, from the above two relations it follows that:

$$|OP| \times |OP'| = |OA| \times |OA'|$$

from where:

$$\frac{|OP|}{|OA|} = \frac{|OA'|}{|OP'|}$$

which precisely means that the lengths of the sides of the two triangles at hand that subtend one and the same interior angle, at the vertex O , are in the same proportion. Which, by Euclid's "*Elements*" Book 6 Proposition 6, also known as **SAS** or side-angle-side, means that these triangles are similar.

Which, in turn, means that the magnitudes of the angles in these triangles that lie opposite corresponding sides must be equal.

From the last relation it follows that:

- the side OP of the right, by construction, triangle OAP corresponds to the side OA' of the triangle $OP'A'$
- the side OA of the triangle OAP corresponds to the side OP' of the triangle $OP'A'$ and
- the side AP of the triangle OAP corresponds to the side $A'P'$ of the triangle $OP'A'$

In the triangle OAP the sides AO and AP subtend the right angle. Hence, in the triangle $OP'A'$ the corresponding sides OP' and $A'P'$ must subtend a right angle as well.

But, since the sample point A was chosen on the given circle, q , at random that precisely means that all the image points A' will belong to a straight line, l , that is perpendicular to the line of centers p .

That straight line l will pass through the common points F_1 and F_2 due to the Theorem 1, because if a sample point A coincides with either of these two points, as in $A = F_1$ or $A = F_2$, then the inversion of the plane at hand will take these points into themselves.

Which exactly means that under the inversion of a plane with respect to a circle $c(O, r)$ with positive power a circle q that passes through the center of inversion, O , and has exactly two distinct points, F_1 and F_2 , common with the circle of inversion, c , is transformed into a straight line, l , that passes through the said common points, F_1 and F_2 . $\square$

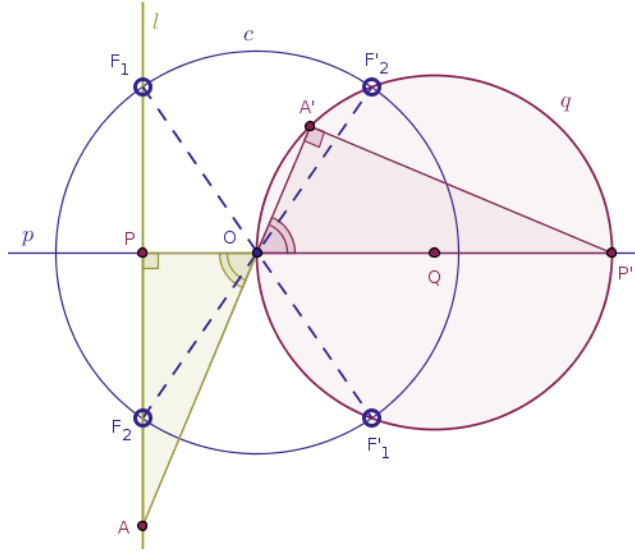


The Euclidean-tools-only construction of the image of the circle from the Theorem 15 is especially pleasant: simply draw a straight line, l , through the two given distinct common points, F_1 and F_2 .

Moving on, the remaining four theorems are proved in a similar way and we suggest the our readers generate all these proofs for practice.

The following theorem is the converse of the Theorem 11.

Theorem 16. Under the inversion of a plane with respect to a circle $c(O, r)$ with negative power a circle q that passes through the center of inversion, O , and has exactly two distinct points, F_1 and F_2 , common with the circle of inversion, c , is transformed into a straight line l , that passes through the points F'_1 and F'_2 which are the points diametrically opposite to the common points F_1 and F_2 correspondingly. For a pictorial demonstration of the Theorem 16 the Figure 5.5 can be reused (Figure 5.5):



In this case, however, the circle q is the preimage figure and the straight line l is the image of the circle q under the inversion of the parent plane with respect to the circle c and as a cosmetic note of no technical significance, the points $F_{1,2}$ and $F'_{1,2}$ should be swapped in the context of the Theorem 16.



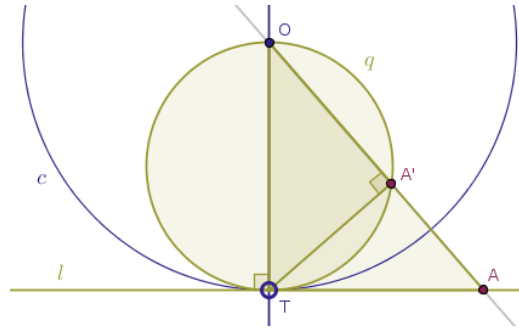
The Euclidean-tools-only construction of the image of the circle from the Theorem 16 is quite pleasant as well: draw two straight lines through the center of inversion, O , and the two given distinct common points, F_1 and F_2 , as shown with dashed lines in Figure 5.5 above, in order to locate the diametrically opposite images of these points, F'_1 and F'_2 , and snap a straight line, l , through these diametrically opposite points F'_1 and F'_2 .

The straight line l thus constructed is the image of the circle from the Theorem 16 sought-after.

The following theorem is the converse of the Theorem 12.

Theorem 17. Under the inversion of a plane with respect to a circle $c(O, r)$ with positive power a circle q that passes through the center of inversion, O , and has exactly one point common with the circle of inversion, c , at a point T is transformed into a straight line, l , that is tangent to the circle of inversion, c , at the tangency point, T .

For a pictorial demonstration of the Theorem 17 the Figure 5.7 can be reused (Figure 5.7):



In the context of the Theorem 17, however, the above circle q is the preimage and the straight line l is the image of the circle q under the corresponding inversion of the parent plane.

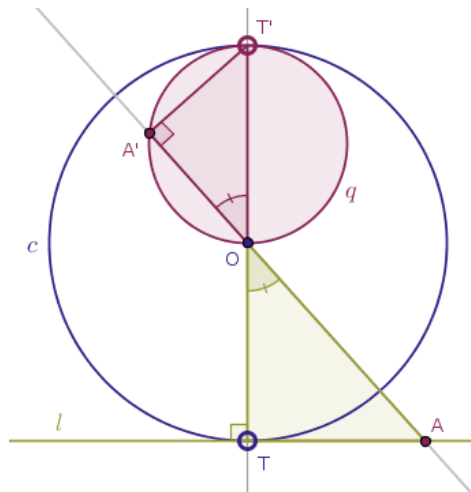


The Euclidean-tools-only construction of the image of the circle from the Theorem 17 is quite pleasant again: draw a straight line, l , tangent to the given circle of inversion, c , through the given point of tangency, T .

The following theorem is the converse of the Theorem 13.

Theorem 18. Under the inversion with respect to a circle $c(O, r)$ with negative power a circle q that passes through the center of inversion, O , and has exactly one point common with the circle of inversion, c , at a point T is transformed into a straight line, l , that is tangent to the circle of inversion, c , at the point T' , where T' is the point diametrically opposite to the tangency point, T .

For a pictorial demonstration of the Theorem 18 the Figure 5.9 can be reused (Figure 5.9):



In the context of the Theorem 18, however, the circle q is the preimage and the straight line l is the image of the circle q and the points T' and T in the figure above should be swapped.



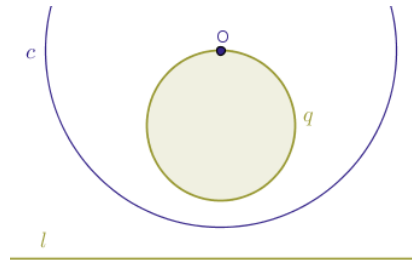
The Euclidean-tools-only construction of the image of the circle from the Theorem 18 is only marginally more complex than the previous construction: draw a straight line through the center of inversion, O , and the given point of tangency, T , in order to locate a diametrically opposite image of that tangency point, T' .

Draw the sought-after straight line, l , tangent to the given circle of inversion at the point T' .

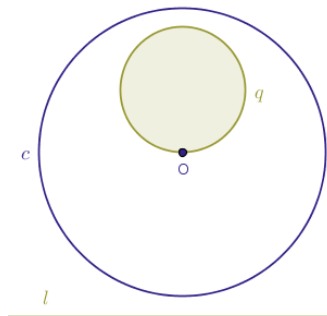
The following theorem is the converse of the Theorem 14.

Theorem 19. Under the inversion of a plane with respect to a circle $c(O, r)$ a circle q that passes through the center of inversion, O , and has no points common with the circle of inversion, c , is transformed into a straight line, l , that has no points common with the circle of inversion, c .

The Theorem 19 holds true for inversions with both powers, positive and negative. The only difference between these powers is the location of the image straight line, l . If the power of inversion is positive (Figure 5.11):



then the image straight line, l , is located on the same side of the center of inversion, O , as the preimage circle, q . And if the power of inversion is negative (Figure 512):



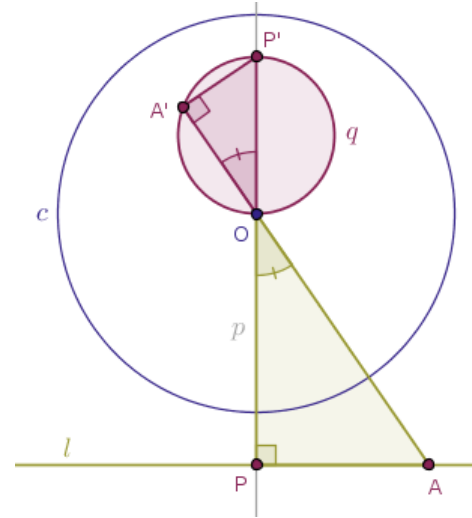
then the image straight line, l , is located on the opposite side of the center of inversion, O , relative to the preimage circle, q .



The Euclidean-tools-only construction of the image of the circle from the Theorem 19 can be lifted almost verbatim from the proof of the Theorem 14. Draw a straight line, p , through the center of inversion, O , and the center of the given circle, Q . The straight line p will intersect the given circle, q , at two distinct points, the center of inversion, O , and its diametrically opposite image, the point P' , as shown in Figure 5.13 on the right.

Reflect the point P' with the given power in the given circle of inversion, c , into a point, named P , as shown on the right.

Draw the straight line sought-after, l , perpendicular to the straight line p through the point P .



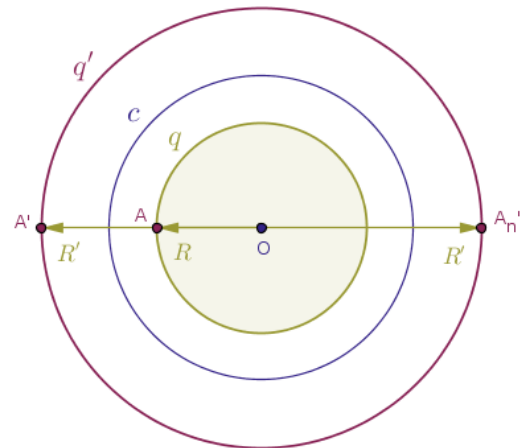
In the remaining theorems of this, and the next, chapter we will study the inversions of circles into circles. Please note carefully here that the center of an original circle does NOT (!) invert into the center of its image directly.

However, there exists an interesting relationship between such two centers and it is possible to invert one into the other not in one but in two steps. Read more on that curious transformation in the Problem 10 in the **Tangency, Power, Radical Axis** chapter of this book.

Theorem 20. Under the inversion of a plane with respect to a circle $c(O, r)$ a circle $q(O, R < r)$ concentric with and located strictly **inside** of the circle of inversion, c , is transformed into a circle $q'(O, R' > r)$ concentric with and located **outside** of the circle of inversion, c .

Proof. Let q be a given circle concentric with and located strictly **inside** of the circle of inversion, c .

An arbitrary original point A on the given circle, q , as shown in the Figure 6.2 on the right, will be taken into its image point A' under the inversion of the parent plane with respect to the circle c with positive power and into its image point A'_n under the inversion of the parent plane with respect to the same circle c with negative power.



Either way, according to Condition 3 of Definition 1, for the above points we must have:

$$|OA| \cdot |OA'| = r^2 = |OA| \cdot |OA'_n|$$

from where for the distances $|OA'|$ and $|OA'_n|$ we have, since $R < r$:

$$|OA'| = |OA'_n| = \frac{r^2}{R} = \text{const} = R' > r$$

Due to the Theorem 4, both image points A' and A'_n must be located **outside** of the circle of inversion, c , and the distances of all such points to the center of inversion, O , will remain constant and strictly larger than the radius of inversion, r . Thus, the locus traced by the image points A' and A'_n is a circle, q' , that is concentric with and is located outside of the circle of inversion, c . $\square$

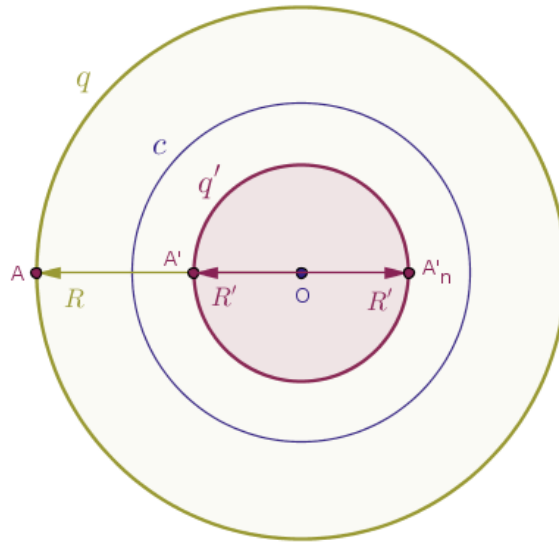


The Euclidean-tools-only construction of the image of the circle from the Theorem 20 is very straightforward: simply reflect any point, A , on the given circle, q , in the given circle of inversion, c , with the given power into its image point, A' . Draw the sought-after circle, q' , with the center of inversion, O , as its center and the distance $|OA'|$ as its radius.

The next theorem is a straightforward converse of the previous theorem.

Theorem 21. Under the inversion of a plane with respect to a circle $c(O, r)$ a circle $q(O, R > r)$ concentric with and located strictly **outside** of the circle of inversion, c , is transformed into a circle $q'(O, R' < r)$ concentric with and located **inside** of the circle of inversion, c .

Proof. Let q be a given circle concentric with and located outside of the circle of inversion, c . An arbitrary original point A on q (Figure 6.3):



will be taken into its image point A' under the inversion with respect to the circle c with positive power and into its image point A'_n under the inversion with respect to the same circle c with negative power.

Either way, according to Condition 3 of Definition 1, for the above points we must have:

$$|OA| \cdot |OA'| = r^2 = |OA| \cdot |OA'_n|$$

from where for the distances $|OA'|$ and $|OA'_n|$ we will have, since $R > r$:

$$|OA'| = |OA'_n| = \frac{r^2}{R} = \text{const} = R' < r$$

Due to the Theorem 5, both image points A' and A'_n must be located strictly inside of the circle of inversion, c , and the distances of all such points from the center of inversion, O , will remain constant. Thus, the locus traced by the image points A' and A'_n is a circle, q' , concentric with and located inside of the circle of inversion, c . $\square$



The Euclidean-tools-only construction of the image of the circle from the Theorem 21 is actually the same as in the previous case: simply reflect any point, A , on the given circle, q , in the given circle of inversion, c , with the given power into its image point, A' . Draw the sought-after circle, q' , with the center of inversion, O , as its center and the distance $|OA'|$ as its radius.

We now prove an important theorem that is used and reused throughout the study of inversive geometry. Because of the importance of the said theorem, we will bestow upon it a special name, **ICSTT**, for **Inversion with respect to a Circle Similar Triangles Theorem**.

Note that the upcoming theorem holds true for both powers of inversion, positive and negative, so long that one and the same power of inversion is used for both pairs of points consistently, either a positive power for all the points at hand or a negative power for all the points at hand.

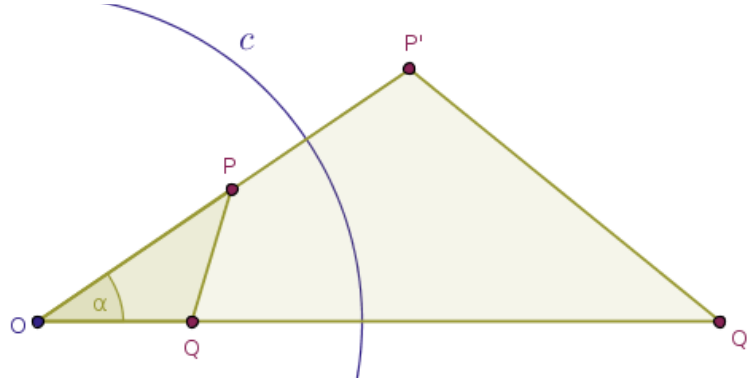
If one power of inversion is used for one pair of points and a different power of inversion is used for the other pair of points then the **ICSTT** is no longer valid.

For demonstration purposes we prove that theorem for a positive power of inversion only and we encourage our readers to adjust that proof for a negative power of inversion on their own.

Theorem 22 (ICSTT). Let the points P and Q in the given plane be distinct from each other and from the points O , the center of inversion, and the point at infinity, P_∞ . Let the points O , P and Q be non-collinear and let the points P' and Q' be the images of the points P and Q under the inversion of a plane with respect to the circle of inversion $c(O, r)$ with positive (negative) power correspondingly.

Then, the triangles whose vertices are ordered as OPQ and $OQ'P'$ are similar.

Proof. Let the premise of our theorem be captured in the following drawing (Figure 6.4):



Observe that the visual presence of the circle of inversion, c , in the diagram above is not technically needed, we show that circle in order to be consistent.

Next, we have already encountered the upcoming argument once, in the proof of the Theorem 10. The reason why we have laid out the relevant proofs in this book in this particular manner, however, is to show to the students of mathematics that the importance of a newly discovered theorem often becomes evident only after the fact, as a deeper study of the subject is undertaken.

Thus, by now it should be a familiar fact that due to how this particular geometric transformation works, with its center of inversion and the collinearity of all the points of interest, by construction, our triangles OPQ and $OQ'P'$ share a common interior angle, α , at their common vertex O . Moreover, by Condition 3 of Definition 1, for each pair of the image points at hand we also must have:

$$|OP| \cdot |OP'| = r^2 = |OQ| \cdot |OQ'|$$

from where it follows that:

$$\frac{|OP|}{|OQ|} = \frac{|OQ'|}{|OP'|}$$

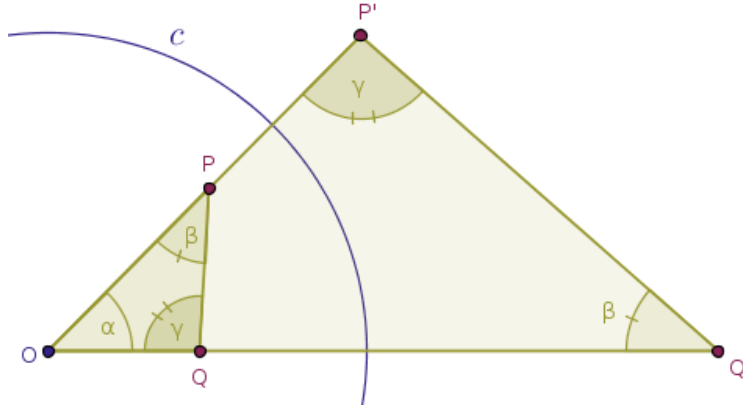
which, in turn, tells us that the sides of our triangles that form the shared angle α are in the same proportion and, thus, by Euclid's "*Elements*" Book 6 Proposition 6, these triangles, namely the triangles OPQ and $OQ'P'$, are similar (by a modern equivalent known as **SAS**, side-angle-side). But in similar triangles equal angles lie opposite proportional sides by the same Book 6 Proposition 5 of Euclid's "*Elements*". Therefore, if we rewrite the last ratio above as follows:

$$\frac{|OP'|}{|OQ|} = \frac{|OQ'|}{|OP|}$$

then we will have no choice but to conclude that the magnitude of the interior angle of the triangle $OQ'P'$ at the vertex Q' that lies opposite the side OP' must be equal to the magnitude of the interior angle of the triangle OPQ at the vertex P that lies opposite to the side OQ :

$$\angle OQ'P' = \angle OPQ = \beta$$

Pictorially (Figure 6.5):



which means that the magnitude of the interior angle of the triangle $OQ'P'$ at the vertex P' that lies opposite the side OQ' must be equal to the magnitude of the interior angle of the triangle OPQ at the vertex Q that lies opposite to the side OP , as in $\angle PQO = \angle Q'P'O = \gamma$, and we now see why the order of the vertices in our triangles is important. $\square$

That is, let us agree that to each planar angle we attach two properties, **the magnitude** and **the direction**. Then, what we just proved is the fact that under any inversion of the real plane with respect to a circle only the said **magnitude** of every relevant angle is preserved, while the said **direction** of every such angle is **not** preserved.

Indeed, the direction of every angle formed by the preimage points and the center of inversion is flipped to its opposite if the corresponding image points are to form a like angle.

For example. The angle OPQ formed by the center of inversion and the two preimage points P and Q will not at all correspond to the angle $OP'Q'$ if we were to simply repeat the names of the vertices of the inverted angle in exactly the same order as the respective preimage points appear in the angle OPQ :

$$\angle OPQ = \beta \neq \gamma = \angle OP'Q'$$

However, we did prove that, because the two triangles at hand are similar, all the relevant pairs of angles in these two triangles have one and the same magnitude:

$$\angle QOP = \angle P'OQ' = \alpha$$

$$\angle OPQ = \angle OQ'P' = \beta$$

$$\angle PQO = \angle Q'P'O = \gamma$$

These relations capture the heart and sole of the fact that while magnitudes of the above pairs angles coincide, their directions are opposite with respect to each other because in all these angles the preimage and the corresponding image points, P and Q versus P' and Q' , are swapped and that is what changes the directions of these angles to their opposite.

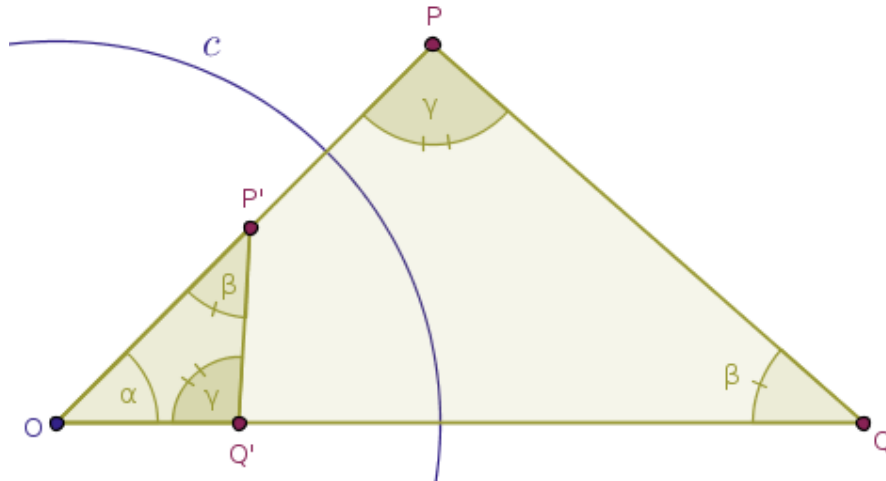
That property of inversion of the real plane in a circle to preserve the magnitudes but to change the directions of all the relevant angles to their opposite is known as **anticonformality**.

In other words, inversion of the real plane in a circle is a mapping (of that plane into itself) that is **anticonformal**.

Separately, note that the above theorem holds true for any mutual juxtaposition of the image points with respect to the circle of inversion:

- if the points P and Q are both located inside of the circle of inversion
- if the points P and Q are both located outside of the circle of inversion and
- if the circle of inversion is located in between the points P and Q

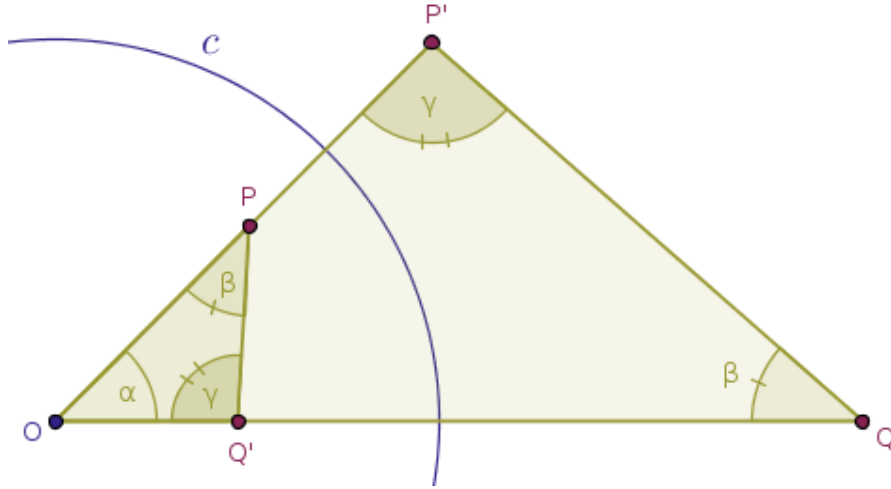
In order to obtain the proofs of the Theorem 22 (ICSTT) for the remaining cases, all we have to do is cyclically permute the names of the vertices of the triangles that have been proven to be similar. For the case when the image points P and Q are both located outside of the circle of inversion, c , we swap P with P' and Q - with Q' so that we have pictorially (Figure 6.6):



and symbolically:

$$\triangle OQP \sim \triangle OP'Q'$$

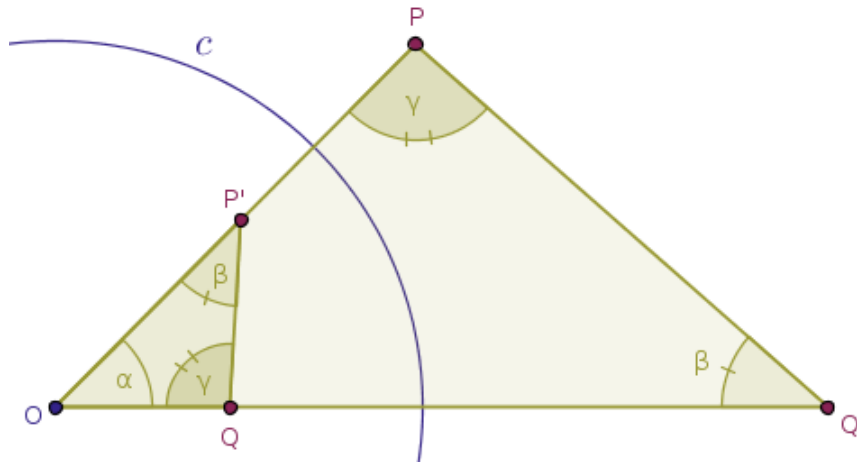
For the case when the point P is located inside of the circle of inversion, c , and the point Q is not, we only swap the points Q and Q' so that we have pictorially (Figure 6.7):



and symbolically:

$$\triangle OPQ' \sim \triangle OQP'$$

For the case when the point Q is located inside of the circle of inversion, c , and the point P is not, we only swap the points P and P' so that we have pictorially (Figure 6.8):



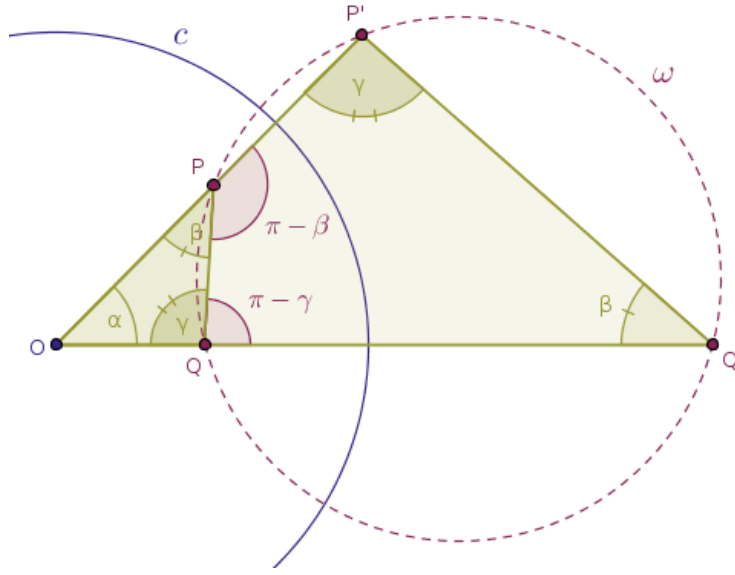
and symbolically:

$$\triangle OP'Q \sim \triangle OQ'P$$

Technically, the proofs of all of these cases are, essentially, the carbon copies of the complete proof given subject to the renaming of the relevant vertices of the relevant triangles.

Corollaries

The Theorem 22 (ICSTT) has a number of interesting and straightforward consequences. For example. If we go back to the drawing shown in Figure 6.5 above then we will notice (Figure 6.9):



that the angles at the vertices Q and P' are supplementary, which instantly implies that the quadrilateral $QQ'P'P$ is cyclic or that all four points of interest are concyclic or that all these points belong to one and the same circle, ω .

If the power of inversion is negative then the instant corollary of the Theorem 22 (ICSTT) is the same but it takes one more step to get there. First, notice that, of course, the triangles OPQ and $OQ'P'$ are similar due to, essentially, the same reasoning that we used in the proof of the Theorem 22.

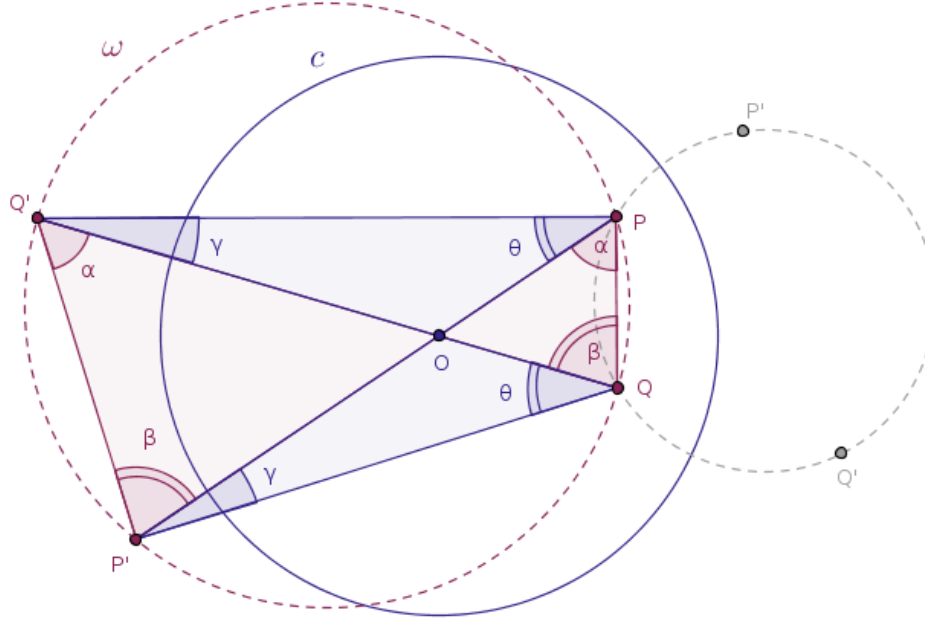
But the main relation between the distances across all four points of interest:

$$|OP| \cdot |OP'| = r^2 = |OQ| \cdot |OQ'|$$

can be carved up in a different way still, as follows:

$$\frac{|OP|}{|OQ'|} = \frac{|OQ|}{|OP'|}$$

which implies that the triangles OPQ' and OQP' are similar as well with the corresponding angles equal one another as shown in the diagram below (Figure 6.10):



But that means that from the triangle PQP' , for example, the magnitudes of all four angles of interest sum to precisely π :

$$\alpha + \beta + \theta + \gamma = \pi$$

which, in turn, implies that, since the magnitudes of the opposite angles in the quadrilateral $PQP'Q'$ sum to π , that quadrilateral is concyclic or that all four points of interest belong to one and the same circle, ω , meaning that all these points are concyclic.

Yet another interesting consequence* of the Theorem 22 (ICSTT) awaits us at the end of the next theorem.

Armed with the Theorem 22 (ICSTT), we can now templatize some number of related proofs of theorems that establish the shape of the image of a given circle that does not pass through the center of inversion.

Theorem 23. Under the inversion of a plane with respect to a circle $c(O, r)$ with positive power a circle, $q(Q, R)$, that does not pass through the center of inversion, O , and has exactly two distinct points, F_1 and F_2 , common with the circle of inversion, c , is transformed into a circle, $q'(Q', R')$, that does not pass through the center of inversion, O , but does pass through the shared points, F_1 and F_2 .

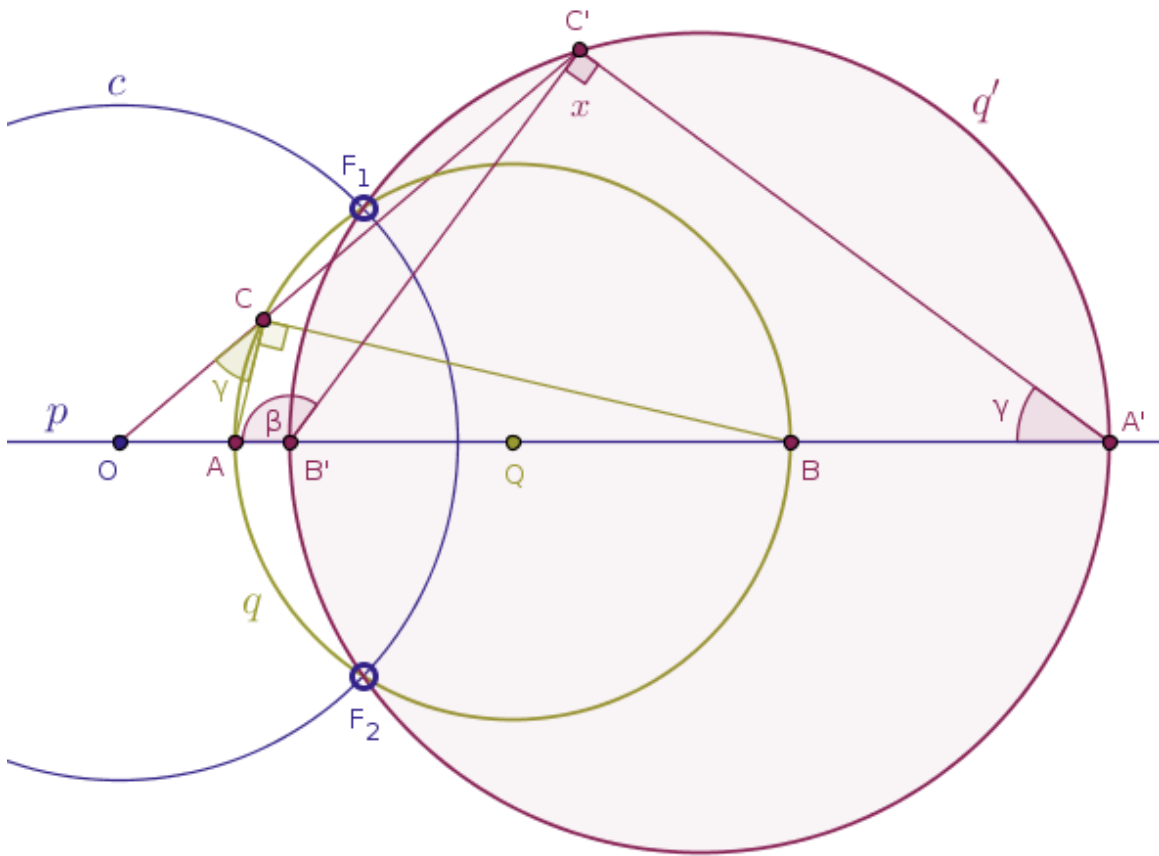
Proof. Let $q(Q, R)$ be a given circle that does not pass through the center of inversion, O , and has exactly two distinct points, F_1 and F_2 , common with the circle of inversion, c .

We observe right away, then, that since the shared points F_1 and F_2 are located on the circle of inversion, c , they, by the Theorem 1, will be fixed and, hence, the curve into which the given circle, q , will be transformed, whatever it is, must pass through these points, F_1 and F_2 .

Next. Draw a straight line, p , through the center of inversion, O , and the center of the given circle, Q , until that straight line intersects the given circle, q , at the points A and B .

Pick a random point, C , on q distinct from the points A , B , F_1 and F_2 . Connect the point C with the points A and B . Reflect the points A , B and C in c with positive power into their corresponding images A' , B' and C' .

By the Theorem 4 and Theorem 5, the image point A' will be located outside of the circle of inversion, c , the image point B' will be located inside of that circle and that circle will be in between the image points C and C' (Figure 6.11):



Since we already know how to cyclically permute the names of the vertices of the triangles similar to the triangles from the Theorem 22 (ICSTT), we right away deduce that the triangles OCA and $OA'C'$ are similar and that it is the case that:

$$\angle OCA = \angle OA'C' = \gamma$$

and that, likewise, the triangles OCB and $OB'C'$ are similar and that it is the case that:

$$\angle OCB = \angle OB'C' = \beta$$

Notice that the angle ACB subtends the diameter of the given circle, q , and, hence, is right:

$$\angle ACB = 90^\circ$$

Next, on the one hand, being composite, the angle OCB is the sum of the following two angles of interest:

$$\angle OCB = \angle OCA + \angle ACB = \gamma + \frac{\pi}{2}$$

On the other hand, the angle $OB'C'$ is exterior to the triangle $B'A'C'$. Hence, by Euclid's "*Elements*" Book 1 Proposition 32, it is the case that:

$$\angle OB'C' = x + \gamma$$

Therefore, by the Theorem 22 (ICSTT), we have:

$$\angle OCB = \gamma + \frac{\pi}{2} = \beta = \angle OB'C' = x + \gamma$$

which is to say that:

$$\gamma + \frac{\pi}{2} = x + \gamma$$

meaning that:

$$x = \frac{\pi}{2} = \angle B'C'A'$$

Therefore, since the original point C was picked at random, it follows that the locus traced by the image points C' must be a circle, q' , that passes through the shared points F_1 and F_2 and that has the line segment $A'B'$ as its diameter.

As such, under the inversion of a plane with respect to a circle $c(O, r)$ with positive power a circle, $q(Q, R)$, that does not pass through the center of inversion, O , and has exactly two distinct points, F_1

and F_2 , common with the circle of inversion, c , is transformed into a circle, $q'(Q', R')$, that does not pass through the center of inversion, O , but does pass through the shared points, F_1 and F_2 , which is what was required to prove. $\square$

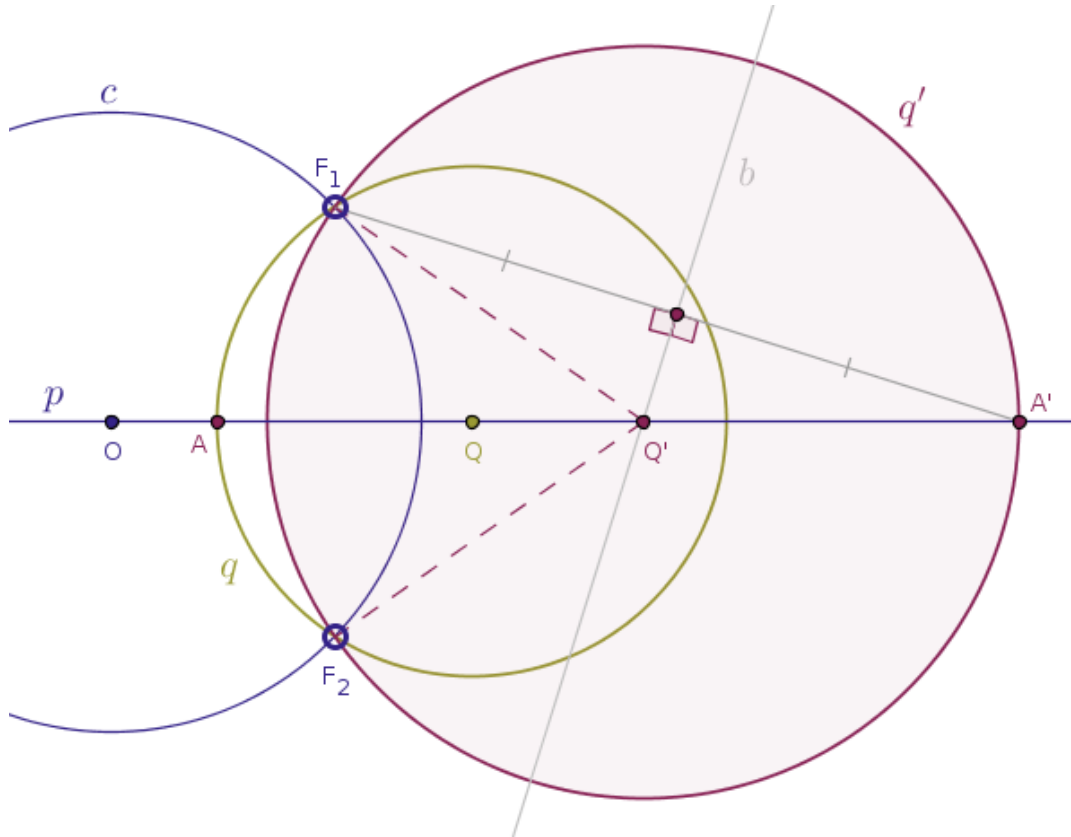
Thus, in order to construct, with Euclidean tools alone, the image of a given circle, $q(Q, R)$, under the inversion of a plane with respect to a circle $c(O, r)$ with positive power when the given circle does not pass through the center of inversion, O , and has exactly two distinct points, F_1 and F_2 , common with the circle of inversion, c , we carry out these steps.

1. Draw a straight line, p , through the center of inversion, O , and the center of the given circle, Q .

That straight line will intersect the given circle, q , at two distinct and diametrically opposite points. Out these two points we need only one. Say, the point that is closest to the center of inversion, O . Name that point A .

2. Reflect that point, A , in the circle of inversion, c , with positive power into its image point, A' .

3. Bisect the line segment $A'F_1$ (or $A'F_2$) with a straight line, b (Figure 6.12):



That perpendicular bisector, b , will intersect the above straight line line of centers, p , at the point Q' .

The point Q' thus constructed is the center of the image circle sought-after, q' .

4. Draw that sought-after circle, q' , centered at the point Q' with the radius $|Q'F_1| = |Q'F_2|$.

A proof of the fact that the above construction steps indeed solve the problem at hand is very straightforward and we will show such a proof for completeness, once, and we will leave the generation of these proofs of the upcoming similar problems as an exercise left to the reader.

The main idea of such a proof is based on the previously proved Theorem 7.

Yet another way to interpret that theorem is as follows. Say, we construct the inversive image, Q , of a point P and after that, once we are done with all the work, we bring a side observer into the room and ask that observer to determine with absolute certainty, across the points P and Q , which point is “the preimage” and which point is “the image”?

Under the respective inversion of the parent plane in a given circle, of course.

You see, due to the symmetry that is built into the inner mechanics of inversion of the real plane in a circle, we claim that there does not exist an algorithm according to which the above question can be answered with absolute certainty.

We, the folks who carried out the actual construction, do know. But the side observer who was not in the room during that construction does not and can not know which point is which.

It is certainly possible to take a wild guess and maybe get the answer to our question right but in the absence of the explicit and the a priori knowledge of which point is the preimage, all that the side observer can do is:

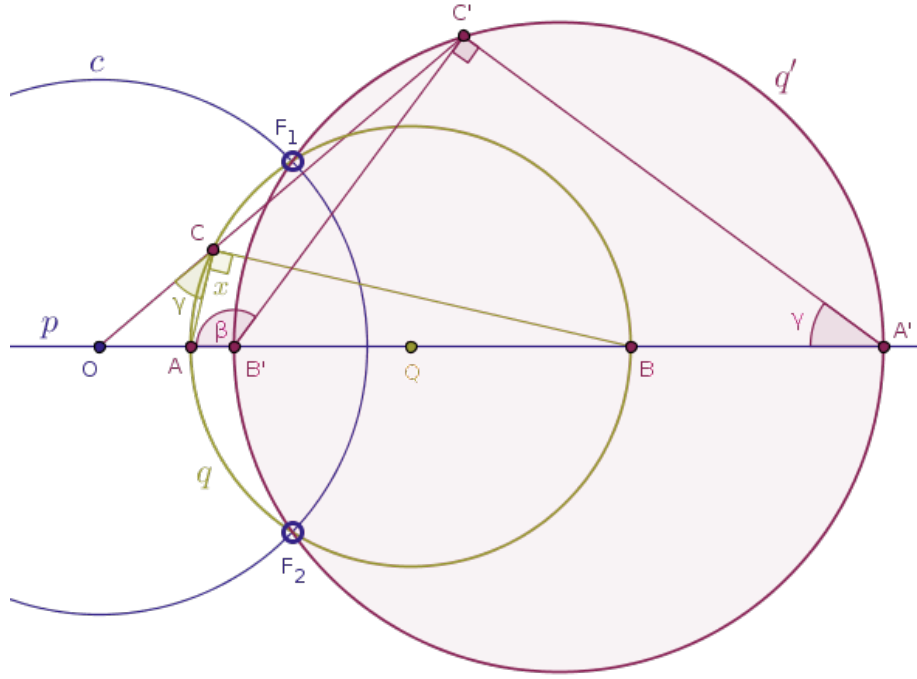
- measure the distances from the center of inversion to the points P and Q
- compute the product of these distances
- witness the fact that that product is equal to the square of the radius of inversion and
- witness the fact that all three points of interest are collinear

And that is all that can be done with certainty. Beyond that the only conclusion that our side observer can draw from this experiment is that the points P and Q are the inversive images of each other.

Which means that any shape that is built up from more than one point will have the same property. In vacuum it is impossible to tell which shape is “the preimage” and which shape is “the image” under the inversion of the real plane with respect to a (given) circle.

That, further, means that in our sample construction above it is impossible to tell the circles q and q' apart when it comes to deciding which such circle is “the preimage” and which such circle is “the image”.

Hence, here we simply take the circle q' to play the role of “the preimage” circle and we show that the orbit that the corresponding points C can trace must be the circle q that has the points A, B as its diameter (Figure 6.13):



And that is more than straightforward to do because we **know** that, since q' is a circle, constructed by us, the magnitude of the interior angle of the triangle $B'C'A'$ and the vertex C' must be right:

$$\angle B'C'A' = \frac{\pi}{2}$$

Because the point C is now “the image” of “the preimage” point C' , by the Theorem 22 (ICSTT), the triangles OCB and $OB'C'$ must be similar with the magnitudes of the interior angles of these triangles at the vertices C and B' respectively equal one another:

$$\angle OCB = \angle OB'C' = \beta$$

But the angle of the triangle OCB at the vertex C is composite, see the diagram above:

$$\angle OCB = \gamma + x$$

where the angle γ is the interior angle of the triangle OCA at the vertex C , which, by the same Theorem 22 (ICSTT) is equal to the magnitude of the angle of the triangle $OA'C'$ at the vertex A' :

$$\angle OCA = \gamma = \angle OA'C'$$

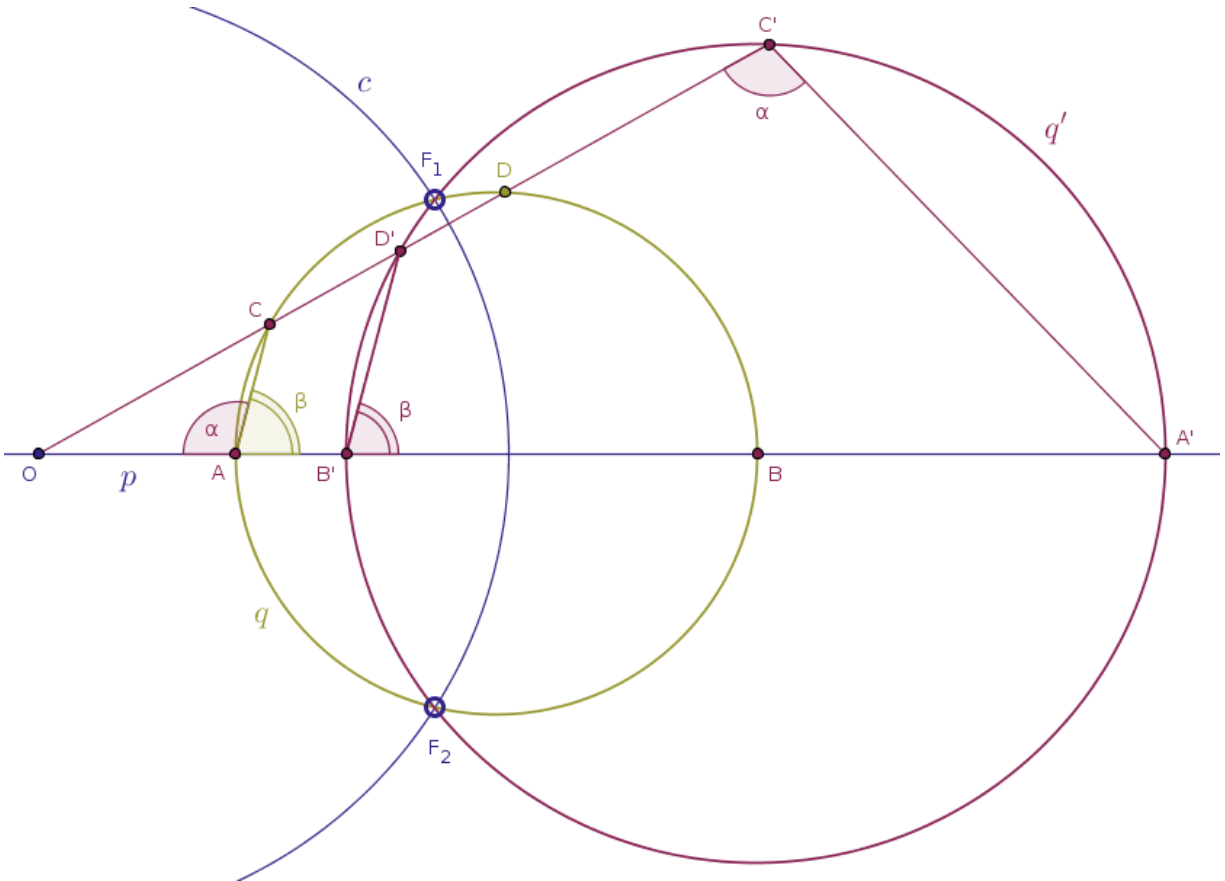
Thus, for the magnitude of the unknown to us angle x , we have:

$$\angle OCB = \gamma + x = \beta = \frac{\pi}{2} + \gamma$$

because the angle β is still an exterior angle of the triangle $BC'A'$. As such, $x = 90^\circ$, which precisely means that the only possible orbit for the image points C is the circle q that has the points A and B as its diameter. $\square$

*Now for the promised interesting and deep consequence of the Theorem 22 (ICSTT). It turns out that if two circles are the images of each other under an inversion of the real plane with respect to a circle centered at a point O with positive (for now) power then that center of inversion O doubles up also as a center of homothety of these two circles.

This statement can be proved as follows. Coming back to our original diagram shown in Figure 11 or in Figure 13, we let two circles q and q' be the images of each other under the inversion of the ambient plane with positive power (Figure 6.14):



Then, observe that the straight line OCC' will intersect the circle q once more at a point, named D , which means that on the same straight line and on the circle q' there will live the image of the point D under the given inversion. Name that image point as D'

Then, based exactly on the Theorem 22 (ICSTT), the pair of points A, C and their inversive images A', C' will form two similar triangles, $\triangle OAC$ and $\triangle OC'A'$, for which the magnitudes of the following interior angles will be equal one another:

$$\angle OAC = \alpha = \angle OC'A'$$

Because the quadrilateral $A'C'D'B'$ is cyclic, by construction, it follows that the magnitudes of its interior opposite angles at B' and at C' must sum to π radians:

$$\angle OC'A' + \angle D'B'A' = \alpha + \beta = \pi$$

Because the angle OAA' is straight or is equal to π radians and is composite, it follows that the magnitudes of the angles on either side of the point A must sum to π radians as well:

$$\angle OAC + \angle CAA' = \alpha + \angle CAA' = \pi$$

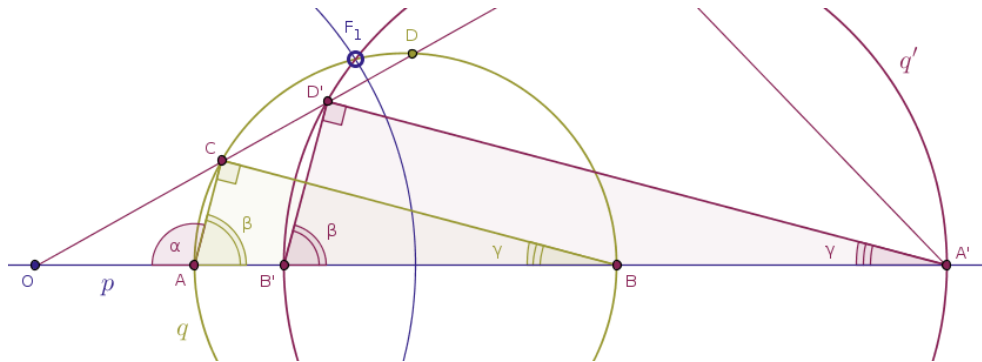
But that means that the sums of the magnitudes of the angles in the last two expressions shown above must be equal one another:

$$\alpha + \beta = \pi = \alpha + \angle CAA'$$

from where it follows that the magnitude of the angle CAA' must be equal to the magnitude of the angle $D'B'A'$:

$$\angle CAA' = \beta = \angle D'B'A'$$

which means that the straight lines AC and $B'D'$ are parallel. Moreover, by construction the angles ACB and $B'D'A'$ subtend the diameters AB and $B'A'$ of the circles q and q' correspondingly (Figure 6.15):



which means that the triangles ACB and $B'D'A'$ are right and, hence, are similar. Which, in turn, means that the straight lines CB and $D'A'$ are parallel.

$$CB \parallel D'A'$$

Since the points A, B', B and A' are collinear, because they all sit on one and the same straight line of centers, the straight line p , and since the points C and D' are collinear, it follows that not only the triangles ACB and $B'D'A'$ are similar but they are also the homothetical images of each other relative to the point O , which, as claimed, doubles up as a center of homothety of the circles q and q' . $\square$

This neat fact can be used for solving various problems but we have to be careful here because the converse of the above theorem is not true in general. That is. In general, two circles have two centers of homothety, an interior center of homothety and an exterior center of homothety.

As such, in general it is not the case that a center of homothety of (any) two circles is also a center of a suitable inversion.

The following theorem, which we will formulate but not prove, clarifies the above situation.

Theorem. If two circles of distinct radii intersect (at two distinct points) then both centers of homothety of these circles can double up as a center of a suitable inversion. But if two circles have no points in common or have precisely one point in common (that is if these two circles touch) then only one center of homothety of two such circles can double up as a center of a suitable inversion.

Here by “inversion” we, of course, understand an inversion of the ambient plane with positive power that takes one of the given circles into the other.